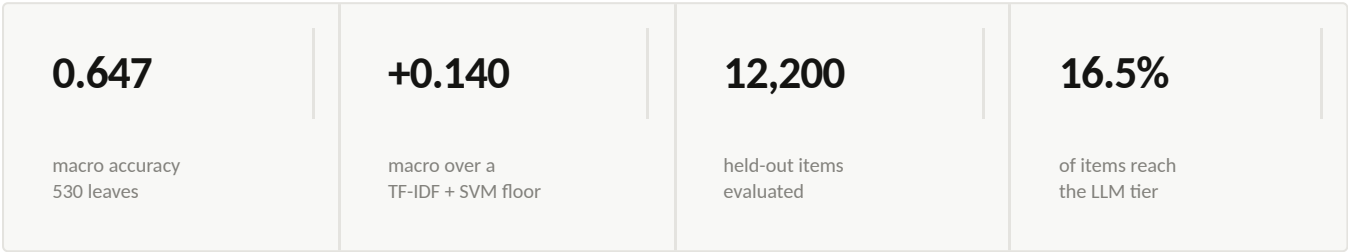


Long-Tail Cascade Classifier

A four-tier classifier for taxonomies where most labels are rare and some have no training data at all. Each tier answers what it can and defers the rest, so cost and capability rise only for the items that need them.



The claim is not that this is a state-of-the-art classifier. It is that every tier has a measured failure attached, and the architecture wins anyway because those failures do not overlap. A TF-IDF + LinearSVC baseline on the same 12,200 items scores micro 0.919 / macro 0.507. The cascade scores **0.895 / 0.647** — it wins macro by 0.140 and loses micro by 0.024. Where that trade comes from is the whole project.

The problem

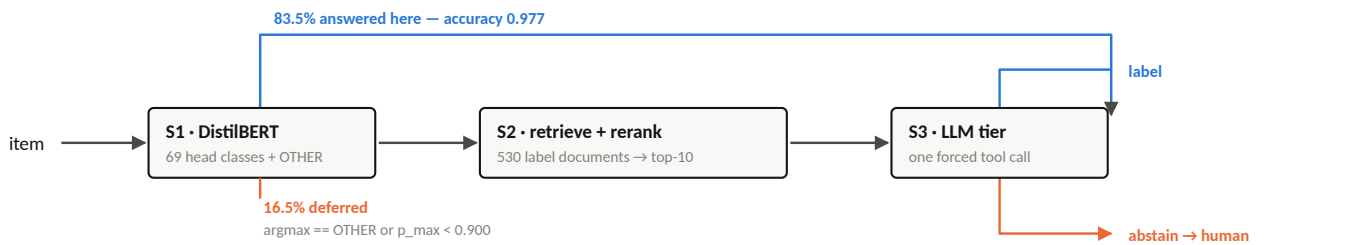
Real taxonomies are Zipfian. In Amazon Berkeley Objects, one leaf is 53.5% of the corpus and 245 leaves have fewer than ten items in it. Two consequences shape everything.

Accuracy is meaningless. Predicting `CELLULAR_PHONE_CASE` for everything scores 53% micro. Every number here is reported micro (per item) and macro (per leaf), and macro is the one to read.

A softmax classifier cannot represent a label it never saw. Not "does badly on" — cannot represent. A flat classifier's score on a genuinely unseen leaf is 0.000 by construction, and no amount of training data for other classes changes it.

The cascade exists for the second point. 90 of the 530 leaves have no training rows at all. Everything else is engineering around that fact.

Architecture



Each tier answers what it can. Cost and capability rise only for the items that need them — and an abstention is a refusal to label, not a fallback guess.

TIER	MODEL	ROLE	SHARE	ACCURACY
S1	DistilBERT, 69 head classes + OTHER	answer the head, defer the rest	83.5%	0.977
S2	frozen Qwen3-Embedding-0.6B + fine-tuned MiniLM cross-encoder	retrieve and rank 10 candidates from 530 (macro recall@10)	—	0.931

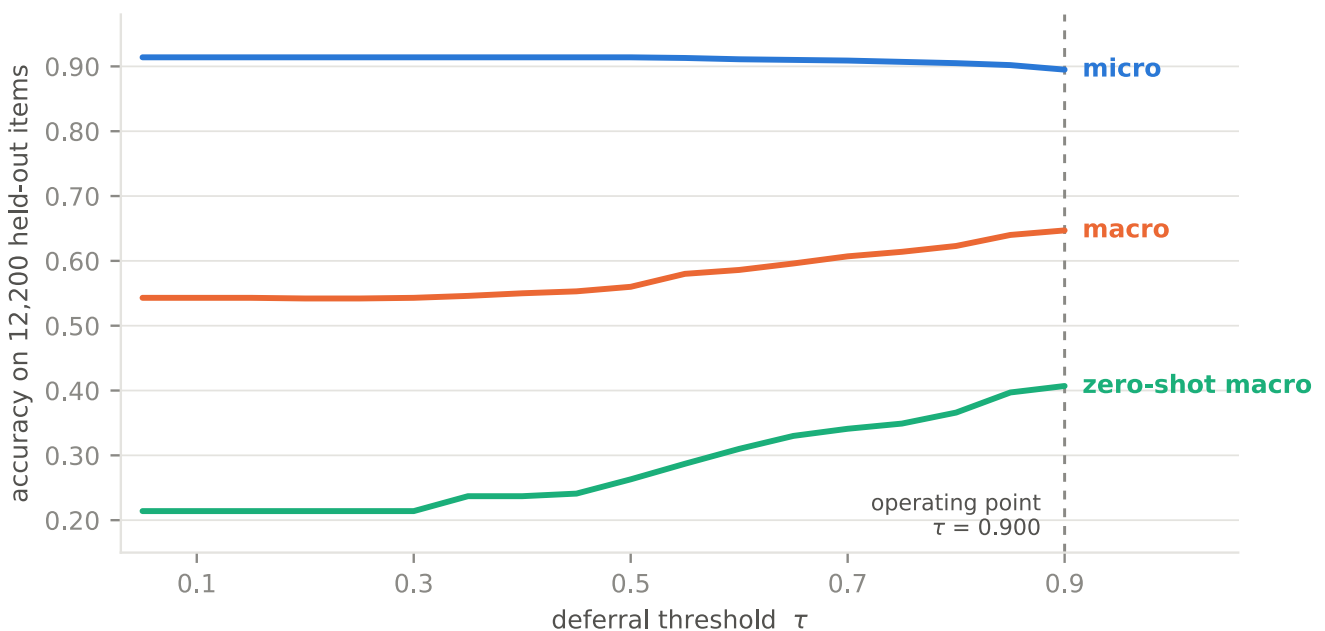
S3	Claude Sonnet 4.5, forced tool call (one call — not an agent loop)	choose from the shortlist, or abstain	16.5%	—
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S1 defers when `argmax == OTHER` or `p_max < 0.900`. The threshold was fitted on a validation split at a 98% precision target, never on test. Retrieval is the ceiling on everything below it: if the correct leaf is not in the top-50, no reranker or prompt can recover it — which is why the agent tier is allowed to *search the taxonomy* rather than only pick from what it was handed.

Prompt engineering as a measurable intervention. A leaf named `ACCESSORY` is unfindable. The same leaf carrying its ancestor path and three real product titles is findable, including at zero training examples. Four cumulative label-document variants, measured on macro recall@50: name **0.837** → + hierarchy path **0.945** → + example products **0.977**.

What the threshold actually buys

The deferral gate is `argmax == OTHER` or `p_max <  $\tau$` , so any τ below the operating point escalates a strict subset of the items already answered on disk. The entire curve below 0.900 is therefore a replay of cached results and costs nothing to produce.



Accuracy against the deferral threshold, replayed from cached Stage 3 answers at zero API cost. Micro falls as macro rises; zero-shot macro nearly doubles. The curve is still climbing where the system operates.

Micro *falls* 0.914 → 0.895 as τ rises while macro *climbs* 0.543 → 0.647, and zero-shot macro nearly doubles, 0.214 → 0.407. That is the project's thesis in one frame: escalating more items trades head-class accuracy for coverage of the tail and of leaves the classifier has never seen. **There is no knee.** Macro is still climbing at the operating point, which means the published τ — chosen for S1's precision in isolation, before the agent tier existed — is not known to be the end-to-end optimum. Finding where macro peaks requires escalating *more*, which is the one direction that costs money.

Findings that generalise

1. Anything trained on the observed label distribution suppresses the unobserved tail

The fine-tuned cross-encoder scores macro recall@10 of **0.760** against the frozen retriever's **0.919** — it is worse than the model it was built to improve. The reason is structural: the 90 zero-training leaves appear in its training groups only as hard negatives, so it is taught they are always wrong, and its zero-shot recall@10 collapses to **0.124**. Its own gate says drop the tier. Interleaving both orderings instead keeps what each is good at: fusing costs 0.006 on few-shot and 0.016 on zero-shot, and buys 0.137 on head and 0.074 on torso.

2. Max-softmax is the wrong instrument for out-of-vocabulary detection

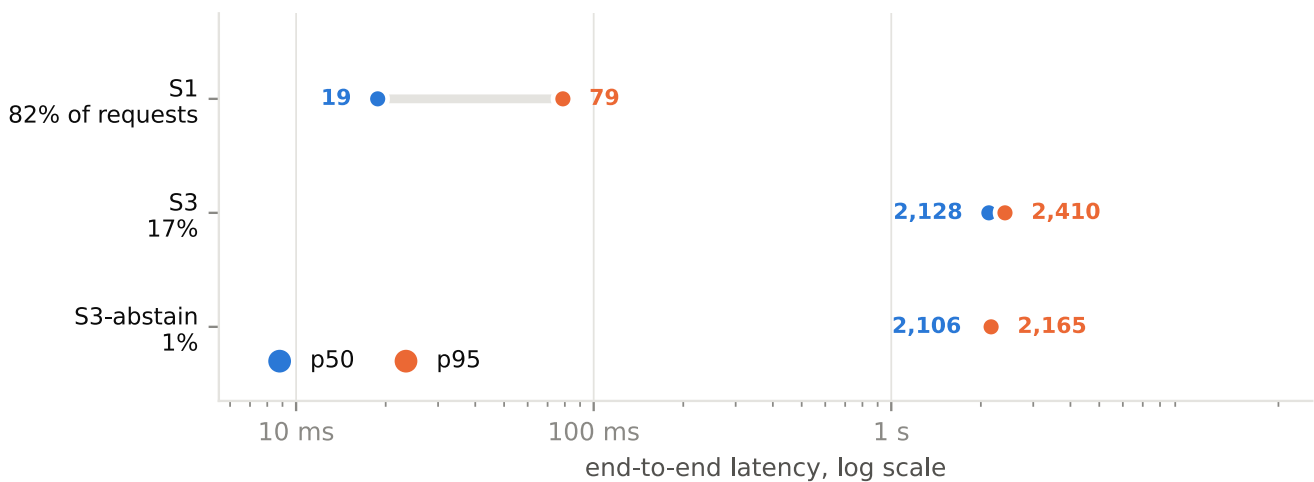
τ moves the confidence bar, but a class the model cannot represent does not produce low confidence — it produces confident wrong answers on a neighbouring class. That is why the gate is a disjunction with an explicit `OTHER` class rather than a threshold alone, and why 875 items defer regardless of τ .

3. The metric that selects a model is not the metric that describes its role

Macro-F1 evaluated the reranker as a classifier and chose the wrong one; the reranker's job is to put the gold leaf in a shortlist, so recall@k is its metric. Selecting on the convenient metric rather than the role-appropriate one is how a tier that hurts the system gets shipped looking like an improvement.

Serving

The cascade runs as a LangGraph state graph behind an HTTP endpoint. Nothing that produces a number is reimplemented — every node calls the same objects the offline evaluation calls, so the served path cannot drift from the measured one. Stage 3 is a swappable provider (hosted API, local model, or cached replay), and 46 assertions cover routing, abstention and the HTTP contract without loading any weights.



200 served requests against a live LLM provider. Cold start excluded and reported separately: the retriever is lazy, so the first deferred request after a restart pays 10.8 s to load it — which 82% of requests never do.

82% of requests are a single DistilBERT forward pass; 17% make a synchronous LLM call. The result is a p50 of 25 ms against a p95 of 2,182 ms — an **87× spread** on a mean of 409 ms that describes no request that was actually made. τ sets the mix, so the accuracy purchase above is also a latency purchase.

Verification, not assumption. Routing in the served path is exact — 0 deferral-decision flips over 400 sampled items. Retrieval is not, and the reason turned out to be a defect in the project's own artefacts: every cached embedding carries an fp16 fingerprint while the classifier probabilities are fp32, meaning they were produced on different machines, and nothing on disk recorded it. It was recovered by testing whether each stored float was exactly representable in half precision. The resulting $\sim 4e-4$ per-vector drift reshuffles a 530-way retrieval enough to cost 2.5 points of gold-leaf reachability, bounding end-to-end impact at **$\leq 1.0\%$ of micro** (95% Wilson). Arrays now carry a provenance sidecar recording resolved device and dtype.

Method: three times measurement beat inference

Each of these began as a confident explanation that was wrong, and ended in a cheap direct measurement that settled it. They are included because the habit is more transferable than any of the results above.

Provenance recovered from bit patterns

The retrieval divergence described above had four plausible causes. Three were eliminated by measurement rather than argument — forcing half precision on the serving side changed nothing, both paths provably build their input with the same function, and the encoder proved batch-invariant to $2.19e-04$. The fourth was found by asking whether each stored float was exactly representable in fp16, which a true fp32 array satisfies with probability $\sim 5e-4$ per element and these satisfied at 1.0000. Arrays now carry a provenance sidecar so the question cannot recur.

The hypothesis that looked certain

Qwen3-Embedding pools the last token, which is only correct under left padding — and the tokenizer's default `padding_side` is `'right'`. That is a textbook silent-corruption bug and it explained the symptom. Measuring the behaviour instead of reading the attribute killed it: the same item encoded alone and in batches of 2, 8 and 16 differed by at most $2.19e-04$, *identical* at every batch size and exactly zero for the longest items. Pad-token pooling would have shown $\sim 1e-1$ and grown with batch size. **The tokenizer attribute is not the behaviour.**

An optimisation that never ran

The LLM tier marks its fixed prompt prefix for caching — 64% of every request is identical across items. Measured against the API's actual rule, the marked prefix was 882 tokens and the model's minimum cacheable prefix is 1,024. Below the minimum the API declines to cache *and returns no error*: both cache counters read zero. Moving the marker to cover the worked examples brought the prefix to 1,282 tokens and made it real — **68% off input cost** in warm steady state, verified from the cache-token fields the response itself reports. An optimisation that fails silently can be believed indefinitely.

The through-line. In all three cases the plausible inference lost to a measurement that took minutes. Two of the numbers in the parity investigation were artefacts of how they were measured before they were findings — a shortlist comparison too strict to return anything but "different," and an alignment keyed on an identifier the dataset duplicates. Both are documented in the repository alongside what the wrong version claimed, because a project that only records its successful measurements is not showing its method.

Honest limitations

- **The cascade loses to a linear SVM on three of five strata.** It wins overall macro because one stratum — zero-shot — is a regime the baseline cannot enter at all.
- **The agent tier is significantly worse than its own shortlist's top candidate on the head stratum.** Measured with a paired McNemar test, not assumed.
- **Abstention targeting is weak:** 7.6% rate, 25.3% recall, 31.4% precision. The largest single pool of recoverable error in the system.
- **Evaluation uses ABO's own labels**, not a human-made gold set, and ABO is a phone-case dataset with a long tail attached — a good tail stress test, a poor proxy for a balanced production taxonomy.
- **Stage 3 is one structured LLM call, not an agent loop.** No tool selection, no multi-turn state, no stopping criterion. The multi-turn version that could search the taxonomy — the only design here able to reach the 9.4% of escalated items whose correct leaf was never retrieved — is specified and not built.
- **A serving path exists; this has not run in production.** No load test, no alerting, no authentication.

PyTorch

Transformers

sentence-transformers

LangGraph

LangChain

LangSmith

FastAPI

Claude API

conformal prediction

McNemar

Corpus: Amazon Berkeley Objects (CC BY-NC 4.0, attribution required, never redistributed). Every figure in this document comes from a run whose artefacts are on disk; the repository documents the measurements behind each decision, including the ones that failed.